Behavioral Expectations*

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Abstract

We study a model where firms form inflation expectations using dual-process reasoning. In familiar states, firms default to a fast, error-prone heuristic (Type 1). In unfamiliar states, they pay a cognitive cost to acquire accurate signals (Type 2). Because large changes in marginal costs push the economy into unfamiliar territory, firms choose to make more accurate expectations precisely when they want to raise prices. The model accounts for three empirical patterns: attention to inflation is high when inflation is also high, belief dispersion co-moves with inflation in a similar fashion, and inflation expectations are the main source of nonlinearity in the Phillips curve. Due to the cognitive costs of high inflation, optimal policy involves greater price stability than under rational expectations.

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1 Introduction

Recent studies of inflation expectations document three empirical patterns. First, attention is state-dependent: individuals pay more attention to inflation when inflation is high (Korenok et al. (2023); Bracha and Tang (2024); Pfäuti (2024); Weber et al. (2025)). Second, cross-sectional belief dispersion co-moves with inflation (Armantier et al. (2025); Kostyshyna et al. (2025); Bullard et al. (2025)). Third, inflation expectations are the main source of nonlinearity in the Phillips curve (Doser et al. (2023); Beaudry et al. (2025); Coibion and Gorodnichenko (2025)). Full-information, rational-expectations models struggle to generate these features.

Instead, these observations suggest that people think harder when the economic environment is unfamiliar, as in the dual-process theories of the cognitive psychology literature (Wason and Evans (1975); Sloman (1996); Kahneman and Frederick (2002); Evans and Stanovich (2013)). These theories provide a useful taxonomy of cognitive processes. Type-1 processes are fast, automatic, and heuristic: they generate default responses at low cognitive cost but are prone to systematic error. Type-2 processes are slow, deliberative, and potentially more accurate, but also more costly. Type 2 tends to override Type 1 when the situation departs sufficiently from what is familiar.

We embed this logic in a stylized general equilibrium model with nominal rigidities. The expectations-augmented Phillips curve is simple: equilibrium inflation is a linear function of a single state variable, x_t, which is perfectly observed by all firms. This state is a composite of the common real marginal cost and average contemporaneous inflation expectations. Firms know that equilibrium inflation is linear in x_t, but do not know the slope, ω. We model this uncertainty as a costly signal extraction problem on ω. In familiar regions of the state space, the payoff to acquiring a signal is small, so firms forgo it and rely on their prior (Type 1). In unfamiliar regions of x_t, forecast errors on the slope translate into large pricing mistakes, and signal acquisition becomes optimal (Type 2). The boundary between the two regimes is determined endogenously in equilibrium.

An increase in real marginal costs has two effects. The standard partial-equilibrium effect is that resetting firms want to raise their prices. The information-acquisition effect is that, because an increase in marginal costs drives x_t to an unfamiliar region, firms are more likely to activate Type 2 and acquire private signals. Firms therefore pay more attention to inflation precisely when inflation is high. Moreover, because when triggering Type 2, firms rely on new draws of private signals, cross-sectional belief dispersion rises after high inflation and contracts when inflation returns to normal.

The individual decision problem of the firm implies that average inflation expectations are

a nonlinear function of real marginal costs. Therefore, the usual Phillips curve relation between inflation and real marginal costs is nonlinear as well. In a "Type 1" region – when marginal costs are small relative to what was experienced before – the slope of the Phillips curve is smaller than under rational expectations: firms retain their prior over the passthrough, inflation expectations are inaccurate, and the response of inflation is muted. In the "Type 2" region – when marginal costs are large – the Phillips curve steepens and converges to the rational-expectations counterpart asymptotically. After a period of high inflation, firms carry better estimates into subsequent periods. Thus inflation becomes more responsive to marginal costs even after inflation has subsided.

In our setup, inflation has two welfare costs that are not present in the canonical New Keynesian model. The first is the cognitive cost of triggering Type 2. The second is the efficiency loss arising from additional price dispersion due to heterogeneous beliefs. Conditional on the same output–inflation trade-off, both costs push the optimal inflation rate below the rational-expectations benchmark.

Our approach is related to three modeling techniques present in literature. The first is rational inattention (Sims (2003); Maćkowiak and Wiederholt (2009); Paciello and Wiederholt (2014)), in which individuals optimally allocate a fixed information-processing capacity across variables. These models struggle to generate attention that is dependent on the realized state. The second approach considers exogenous departures from rational expectations (Marcet and Nicolini (2003); Gáti (2023); Pfäuti (2024)). These models fit survey data on expectation formation well, but because the formation process is specified exogenously, it is not invariant to policy. The third approach studies endogenous expectation formation (Afrouzi et al. (2024); Baley and Turen (2025); Turen (2023)). Afrouzi et al. (2024) study pricing and information frictions in the context of monetary non-neutrality; Baley and Turen (2025) model lumpy forecast revisions by professional forecasters; Turen (2023) analyzes state-dependent attention and pricing decisions in partial equilibrium. These papers do not derive implications for the shape of the Phillips curve and for optimal monetary policy.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 characterizes the Phillips curve. Section 5 studies optimal policy. Section 6 concludes.

2 Related Literature

2.1 Dual-Process Reasoning

The expectation formation process in this paper builds on dual-process theories from cognitive psychology, which distinguish two types of cognitive processes (Evans (2008); Evans and Stanovich (2013)). Type 1 processes are fast, automatic, and low-effort; they produce default responses that are often adequate but prone to systematic error. Type 2 processes are slow, deliberative, and demanding of working memory; they can monitor Type 1 output and override it when warranted. Classic experiments, such as the Wason selection task (Wason and Evans (1975)), or the bat-and-ball problem (Kahneman and Frederick (2002)), illustrate the distinction. In both cases, the task has a clear logical structure, yet most people default to a fast heuristic response and fail to engage the deliberative reasoning required to reach the correct answer. Evans and Stanovich (2013) articulate the default-interventionist position: most behavior accords with Type 1 defaults, and Type 2 intervention occurs only when difficulty, novelty, and motivation combine to command working-memory resources. Thompson et al. (2011) and Thompson et al. (2013) provide direct support, showing that people who answer more quickly when pressed for time are less likely to revise their answer. Stanovich and West (2000) demonstrate that, conditional on Type 2 being triggered, cognitive ability predicts the probability of successful override.

2.2 Empirical Evidence

Several studies document that attention to inflation is state-dependent. Korenok et al. (2023) show that internet search intensity for "inflation" jumps sharply once inflation exceeds a threshold near 4 percent. Bracha and Tang (2024) find that consumer inattention falls dramatically above similar thresholds in the Michigan Survey of Consumers. Pfäuti (2024) estimates an attention threshold at roughly 4 percent. Weber et al. (2025) use randomized controlled trials across countries to show that agents in high-inflation environments become more attentive and less responsive to exogenously provided information.

Cross-sectional disagreement about inflation also varies with the inflation environment. Armantier et al. (2025) and Kostyshyna et al. (2025) document that heterogeneity in inflation expectations rose sharply during the post-2021 inflation surge. Bullard et al. (2025) find that dispersion increased during the surge and declined during the disinflation.

A parallel literature examines the shape of the Phillips curve. Doser et al. (2023) show that its apparent nonlinearity diminishes substantially once inflation expectations are controlled for.

Beaudry et al. (2025) confirm that evidence for a nonlinear Phillips curve is fragile and largely attributable to expectation formation. Coibion and Gorodnichenko (2025) review the broader evidence on inflation expectations and monetary policy.

Malmendier and Nagel (2016) show that individuals who have lived through higher inflation hold persistently higher expectations, with more recent experiences receiving greater weight. Their experience-based learning model captures cross-cohort heterogeneity but, as Gennaioli et al. (2025) demonstrate, cannot account for the rapid de-anchoring during the post-2021 surge nor for the stronger initial reaction by older cohorts. Gennaioli et al. (2025) extend the role of experiences by modeling selective memory: current inflation acts as a numerical cue that triggers retrieval of similar past episodes. Their mechanism operates through recall rather than deliberation.

2.3 Theoretical Literature

The modeling approach builds on Ilut and Valchev (2023), who formalize economic agents as imperfect problem solvers facing uncertainty about an unknown optimal policy function and acquiring signals subject to cognitive costs. We apply their method to the special case of a linear forecasting function; the specialization yields closed-form characterizations of the Phillips curve and of optimal monetary policy not available in the general framework. Rebelo et al. (2025) also build on Ilut and Valchev (2023), applying dual-process reasoning to the consumer side: monopolistic firms influence whether households use System 1 or System 2 through pricing decisions, generating a novel source of endogenous price rigidity. Gabaix and Graeber (2024) and Gabaix (2025) study related models in which individuals have uncertainty about the optimal response to state variables. Bordalo et al. (2025) develop a cognitive theory in which attention to features of the current problem is determined by the decision maker's categorization within previously encountered problems, generating heterogeneity in attention, rigidity when categorization does not change, and discontinuous shifts upon re-categorization.

In the rational inattention framework of Sims (2003), agents face a constraint on information-processing capacity measured by Shannon mutual information. Maćkowiak and Wiederholt (2009) show that rationally inattentive firms allocate more capacity to higher-variance variables. Paciello and Wiederholt (2014) derive optimal monetary policy when firms' attention is endogenous but determined before shocks are realized. Turen (2023) extends the framework to fully dynamic and flexible information acquisition in pricing, showing that state-dependent attention can reconcile micro price data with evidence on information acquisition over the business cycle. **?** introduce smooth diagnostic expectations, under which agents over-react to news

relative to a reference information set and the magnitude of over-reaction varies systematically with uncertainty. In all of these models, the amount of information acquired depends on the variance of shocks, not on the level of the state. Our model makes a different prediction: a large realization of marginal costs triggers learning; a small one does not.

A large literature departs from rational expectations by specifying the forecasting model exogenously. Evans and Honkapohja (2001) provide a comprehensive treatment of adaptive learning. Marcet and Nicolini (2003) show that occasional switches to constant-gain learning account for recurrent hyperinflations. Gáti (2023), Pfäuti (2024), and Branch (2004) propose models with state-dependent forecasting rules that do not emerge from an optimization problem. The expectation formation process in our model is the solution to a well-defined decision problem and is therefore invariant to the monetary policy regime.

Afrouzi et al. (2024) study the joint contribution of pricing and information frictions to the non-neutrality of money, focusing on decomposing sources of non-neutrality rather than on the shape of the Phillips curve or optimal policy. Baley and Turen (2025) model lumpy forecast revisions by professional forecasters, combining Bayesian updating with fixed revision costs and strategic complementarities; their analysis is partial-equilibrium and designed to match cross-sectional survey properties. Molavi (2019) proposes a general theory of equilibrium with model misspecification. The present paper derives specific predictions for the Phillips curve and optimal monetary policy from a general-equilibrium model grounded in dual-process reasoning.

3 The Setup

Time is discrete and indexed by $t = 0, 1, \ldots$ The economy is populated by a continuum of measure one of firms indexed by $i \in [0, 1]$. Each firm consists of a forecaster and a price setter (Appendix C microfounds the forecaster's payoff from the household's problem). The profit of each firm in period t is

$$-\frac{\vartheta}{2}\left(p_{i,t} - p_t^*\right)^2,$$

where $p_{i,t}$ is the nominal price set by firm i, and p_t^* is the ideal nominal price, assumed to be common across firms. The ideal price is given by

$$p_t^* = p_t + \mathrm{mc}_t,$$

where

$$p_t = \int_0^1 p_{i,t} di \tag{1}$$

is the price level in period t, and mc_t is the real marginal cost.

At the beginning of period t, mc_t is revealed and all forecasters form expectations about inflation. A fraction ω of firms is then allowed to change prices. The resetting firms choose their price before observing the nominal prices set by other firms. The reset price is

$$p_{i,t} = \mathbb{E}_{i,t} p_t + \mathrm{mc}_t, \;\; i \in \mathcal{R}_t \tag{2}$$

where $\mathbb{E}_{i,t}$ is the expectation of forecaster i and $\mathcal{R}_t$ is the set of resetting firms, while the sticky-price firms set

$$p_{i,t} = p_{i,t-1}, \;\; i \notin \mathcal{R}_t. \tag{3}$$

Equations (1)-(3) imply the expectations-augmented Phillips curve

$$\pi_t = \omega \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t \right), \tag{4}$$

where $\overline{\mathbb{E}}_t \pi_t \equiv \int_0^1 \mathbb{E}_{i,t} \pi_t \, di$ is the average expectation across all forecasters. Because the Calvo lottery is i.i.d. across firms, the law of large numbers ensures that the average expectation of resetting firms equals the population average.

Equation (2) implies that, when making their pricing decisions, firms do not take into account the fact that they might not be able to change prices in the future. This assumption preserves the simplicity of the equilibrium relation (4) and shifts the focus of the analysis to expectation formation.

3.1 Expectation Formation

Under rationality, forecasters understand equation (4) and set

$$\mathbb{E}_{i,t} \pi_t = \omega \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t \right).$$

Aggregating and solving for $\overline{\mathbb{E}}_t \pi_t$,

$$\overline{\mathbb{E}}_t \pi_t = \frac{\omega}{1-\omega} \mathrm{mc}_t = \pi_t.$$

Holding expectations fixed, a one-unit increase in mc_t raises π_t by ω. Under rational expectations, forecasters understand the mapping, revise expected inflation accordingly, and inflation feeds back into itself through expectations. Iteration gives $\omega + \omega^2 + \omega^3 + \cdots = \omega / (1-\omega)$. Thus, $1/(1-\omega)$ is the rational-expectations feedback multiplier.

We depart from rational expectations by assuming that forecasters have a limited understanding of equation (4). While forecasters understand that equilibrium inflation is a linear function of $x_t \equiv \text{mc}_t + \overline{\mathbb{E}}_t \pi_t$, they are uncertain about the passthrough from x_t to π_t. That is, each forecaster believes that inflation is determined according to

$$\pi_t = f(x_t) = \hat{\omega} x_t,$$

where $\hat{\omega}$ is a random variable. Each forecaster then solves a costly signal extraction problem on $f(x_t)$.

At the beginning of period t, forecaster i's prior belief over $\hat{\omega}$ is

$$\hat{\omega} \mid \mathscr{I}_{i,t-1} \sim \mathcal{N}\left(\mu_{i,t-1}, \sigma^2_{i,t-1}\right),$$

where $\mathscr{I}_{i,t-1}$ is forecaster i's information set at the beginning of period t. Each forecaster can extract a costly normal signal on $f(x_t)$ with perceived distribution

$$s_{i,t} = \hat{\omega} x_t + \sigma_{\epsilon,i,t} \epsilon_{i,t}, \;\; \epsilon_{i,t} \sim \mathcal{N}(0,1).$$

The intra-period timing of information acquisition is:

1. x_t is realized, and each forecaster i chooses $\sigma_{\epsilon,i,t}$;

2. $s^*_{i,t} = \omega x_t + \sigma_{\epsilon,i,t} \epsilon_{i,t}$ is realized, and the prior moments are updated according to Bayes' law to form the posterior belief

$$\hat{\omega} \mid \mathscr{I}_{i,t} \sim \mathcal{N}\left(\mu_{i,t}, \sigma^2_{i,t}\right),$$

   where[1]

$$\mu_{i,t} = \mu_{i,t-1} + \frac{\sigma^2_{i,t-1} x_t}{\sigma^2_{i,t-1} x_t^2 + \sigma^2_{\epsilon,i,t}} \left(\omega x_t + \sigma_{\epsilon,i,t} \epsilon_{i,t} - \mu_{i,t-1} x_t\right),$$

   and

$$\sigma^2_{i,t} = \frac{\sigma^2_{i,t-1} \sigma^2_{\epsilon,i,t}}{\sigma^2_{i,t-1} x_t^2 + \sigma^2_{\epsilon,i,t}}. \tag{5}$$

3. Each forecaster i sets the expectation $\mathbb{E}_{i,t} \pi_t$;

[1] These updating formulas are the standard conditional-normal expressions for the posterior mean and variance.

4. π_t is realized, and forecaster i obtains payoff

$$v_{i,t} = -\frac{\vartheta}{2}\left(\mathbb{E}_{i,t}\pi_t - \pi_t\right)^2 - \mathcal{I}_{i,t},$$

where $\mathcal{I}_{i,t}$ is proportional to Shannon mutual information,

$$\mathcal{I}_{i,t} = \frac{\iota}{2}\ln\left(\frac{\sigma^2_{i,t-1}}{\sigma^2_{i,t}}\right), \quad \iota > 0.$$

Note that forecasters perceive the signal law as $s_{i,t} = \hat{\omega}x_t + \sigma_{\epsilon,i,t}\epsilon_{i,t}$, with uncertainty over $\hat{\omega}$. The true passthrough is ω, so the true law is $s^*_{i,t} = \omega x_t + \sigma_{\epsilon,i,t}\epsilon_{i,t}$.

The problem of forecaster i is

$$\max_{\sigma_{\epsilon,i,t}\geq 0} -\frac{\vartheta}{2}\min_{\mathbb{E}_{i,t}\pi_t}\mathbb{E}\left[\left(\mathbb{E}_{i,t}\pi_t - \hat{\omega}x_t\right)^2 \mid \mathscr{I}_{i,t}\right] - \frac{\iota}{2}\ln\left(\frac{\sigma^2_{i,t-1}}{\sigma^2_{i,t}}\right), \tag{6}$$

where the posterior variance is given by equation (5). We split the problem of the forecaster in two parts:

1. Given $\sigma_{\epsilon,i,t}$, choose the optimal expectation $\mathbb{E}_{i,t}\pi_t$;

2. Given $\mathbb{E}_{i,t}\pi_t$, optimally choose $\sigma_{\epsilon,i,t}$.

(6) shows that cognitive costs do not depend on $\mathbb{E}_{i,t}\pi_t$. Given $\sigma_{\epsilon,i,t}$, the problem of choosing the optimal expectation is

$$\min_{\mathbb{E}_{i,t}\pi_t}\mathbb{E}\left[\left(\mathbb{E}_{i,t}\pi_t - \hat{\omega}x_t\right)^2 \mid \mathscr{I}_{i,t}\right]. \tag{7}$$

The optimal expectation is equal to the posterior mean,

$$\mathbb{E}_{i,t}\left[\pi_t\right] = \mu_{i,t}x_t, \tag{8}$$

Substituting (8) into (6), the problem of choosing the signal variance becomes

$$\max_{\sigma^2_{i,t}\leq\sigma^2_{i,t-1}} -\frac{\vartheta}{2}\min_{\mathbb{E}_{i,t}\pi_t}\sigma^2_{i,t}x_t^2 - \frac{\iota}{2}\ln\left(\frac{\sigma^2_{i,t-1}}{\sigma^2_{i,t}}\right).$$

This problem can be solved in terms of the posterior variance, $\sigma^2_{i,t}$, instead of the signal noise variance, $\sigma^2_{\epsilon,i,t}$. The constraint $\sigma^2_{i,t} \leq \sigma^2_{i,t-1}$ ensures that the solution is consistent with Bayes' rule.

Taking the first-order condition and imposing the constraint, the optimal solution is

$$\sigma_{i,t}^2 = \min\left\{\frac{\overline{\sigma}^2}{x_t^2}; \sigma_{i,t-1}^2\right\} = \begin{cases} \frac{\overline{\sigma}^2}{x_t^2}, & \text{if } x_t^2 > \frac{\overline{\sigma}^2}{\sigma_{i,t-1}^2} \\ \sigma_{i,t-1}^2, & \text{if } x_t^2 \leq \frac{\overline{\sigma}^2}{\sigma_{i,t-1}^2} \end{cases}. \tag{9}$$

where $\overline{\sigma}^2 \equiv \iota/\vartheta$.

Equation (9) illustrates how the forecasters in our model employ dual-process reasoning to form their expectations. When x_t^2 is small, prior uncertainty $\sigma_{i,t-1}^2 x_t^2$ is low. Forecasters optimally choose not to draw a new signal, and rely on their prior belief (Type 1). When x_t^2 crosses the threshold $\overline{\sigma}^2/\sigma_{i,t-1}^2$, prior uncertainty is high. Forecasters draw a new signal and update their beliefs (Type 2). The threshold increases if (1) the weight on the cognitive cost is high relative to the weight on the forecast error, or (2) prior variance is low. Thus forecasters choose to draw a signal only when they are not confident about their assessment.

Under optimal information acquisition, we can write the posterior moments in terms of a reference value for x_t. Let

$$x_{i,r,t}^2 \equiv \frac{\overline{\sigma}^2}{\sigma_{i,t}^2}$$

Lemma 1 describes the posterior moments under optimal information acquisition in terms of the reference value $x_{i,r,t}$.

Lemma 1. *The posterior moments obey the recursion*

$$\mu_{i,t} = \mu_{i,t-1} + \alpha_{i,t}\left(\omega - \mu_{i,t-1}\right) + sign\left(x_t\right)\sqrt{\frac{\alpha_{i,t}\left(1-\alpha_{i,t}\right)}{x_{i,r,t-1}^2}}\overline{\sigma}\epsilon_{i,t}, \tag{10}$$

$$\alpha_{i,t} = \max\left\{0; 1 - \frac{x_{i,r,t-1}^2}{x_t^2}\right\}, \tag{11}$$

$$x_{i,r,t} = \begin{cases} x_t, & \text{if } x_t^2 > x_{i,r,t-1}^2 \\ x_{i,r,t-1}, & \text{if } x_t^2 \leq x_{i,r,t-1}^2 \end{cases}, \tag{12}$$

$$\sigma_{i,t}^2 = \frac{\overline{\sigma}^2}{x_{i,r,t}^2},$$

given $\left(\mu_{i,-1}, x_{i,r,-1}\right)_{i\in[0,1]}$.

Proof. See Appendix A.1. □

3.2 Equilibrium in Price Setting

Together with equation (4), the expectation formation process described in the previous section completes the description of an equilibrium in price setting, which characterizes the mapping from the sequence of exogenous real marginal costs to the sequence of inflation rates.

Definition 1 (Equilibrium in Price Setting)**.** An equilibrium in price setting is a sequence

$$\left\{\pi_t, \left(\mu_{i,t}, x_{i,r,t}, \alpha_{i,t}\right)_{i\in[0,1]}\right\}_{t=0}^{\infty}$$

such that, given the sequence $\{\mathrm{mc}_t\}_{t=0}^{\infty}$, and initial conditions $\left(\mu_{i,-1}, x_{i,r,-1}\right)_{i\in[0,1]}$, the following are satisfied:

$$\begin{aligned} \pi_t &= \omega\left(\mathrm{mc}_t + \overline{\mathbb{E}}_t\pi_t\right), \\ \overline{\mathbb{E}}_t\pi_t &= \overline{\mu}_t\left(\mathrm{mc}_t + \overline{\mathbb{E}}_t\pi_t\right), \\ \overline{\mu}_t &= \int_0^1 \mu_{i,t}\, di, \end{aligned} \tag{13}$$

and $\mu_{i,t}$, $x_{i,r,t}$ and $\alpha_{i,t}$ follow the recursions (10)-(12).

Equation (13) characterizes *equilibrium expectations*. The solution to this equation consists of the average expectation that is consistent with optimality of individual expectations. This fixed point differs from the rational expectations equilibrium due to the fact that $\overline{\mu}_t \neq \omega$ in general.

4 Phillips Curve

In this section we study the properties of the equilibrium in price setting. We begin by imposing the following assumption.

Assumption 1. $x_{i,r,-1}^2 = x_{r,-1}^2$ *for all* i *and* $\overline{\mu}_{-1} = 0$.

We assume that the initial reference state is common across forecasters. This assumption is consistent with Pfäuti (2024), who estimates that individuals have a threshold for attention to inflation of 4%.

We interpret the assumption of $\overline{\mu}_{-1} = 0$ as a partial equilibrium prior. Initially, forecasters on average do not understand the effect of marginal costs on aggregate inflation.

This assumption implies that

$$x_{i,r,t} = x_{r,t}, \;\; t \geq 0,$$

$$\alpha_{i,t} = \alpha_t, \;\; t \geq 0,$$

and

$$\mu_{i,t} < \omega, \;\; t \geq 0.$$

The average mean belief evolves as

$$\overline{\mu}_t = \overline{\mu}_{t-1} + \alpha_t \left(\omega - \overline{\mu}_{t-1}\right), \tag{14}$$

where

$$\alpha_t = \max\left\{0; 1 - \frac{x_{r,t}^2}{x_t^2}\right\}. \tag{15}$$

Equations (14) and (15) illustrate that forecasters update their beliefs according to a state-dependent adaptive expectation process. This process differs from Gáti (2023) due to the fact that the endogenous gain, α_t, is the result of an optimization problem. When $x_t^2 \leq x_{r,t}^2$, forecasters do not learn. When $x_t^2 > x_{r,t}^2$, forecasters trigger Type 2. As $x_t^2 \to \infty$, forecasters exert increasingly more cognitive effort and their beliefs converge to rational expectations (on average).

Under the previous assumption, we can use equation (13) to compute equilibrium expectations in closed-form. The following theorem uses that expression to define the Phillips curve in this economy.

Theorem 1. *The Phillips curve is*

$$\mathcal{PC}_t\left(mc_t\right) = \omega \mathcal{X}_t\left(mc_t\right),$$

where

$$\mathcal{X}_t\left(mc_t\right) = \begin{cases} \frac{mc_t - \sqrt{mc_t^2 - \Delta_t}}{2(1-\omega)} & mc_t < -\ell_t \\ \frac{1}{1-\overline{\mu}_{t-1}} mc_t & -\ell_t \leq mc_t \leq \ell_t \\ \frac{mc_t + \sqrt{mc_t^2 - \Delta_t}}{2(1-\omega)} & mc_t \geq \ell_t \end{cases}, \tag{16}$$

$$\Delta_t \equiv 4\left(1-\omega\right)\left(\omega - \overline{\mu}_{t-1}\right) x_{r,t-1}^2,$$

and

$$\ell_t \equiv \left(1 - \overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right|.$$

Proof. See Appendix A.2. □

Theorem 1 shows that the Phillips curve has two regions: the Type-1 region where $|\text{mc}_t| \leq \ell_t$, and the Type-2 region where $|\text{mc}_t| > \ell_t$. In the Type-1 region the Phillips curve is linear, and

flatter than under rational expectations. In the Type-2 region, the Phillips curve is nonlinear and converges to rational expectations as $|\mathrm{mc}_t| \to \infty$.

This theorem also shows that the slope of the Phillips curve is affected by experience. When forecasters trigger Type 2 in period t, the slope of the Phillips curve in the Type-1 region of period $t+1$ becomes closer to ω. Furthermore, the boundary of the Type-1 region increases. Therefore, learning at t implies that learning region at $t+1$ is smaller, but that conjectures are more precise. This discussion is summarized in the following theorem.

Theorem 2. *Let*

$$\mathcal{X}^*(mc_t) = \frac{1}{1-\omega} mc_t,$$

the equilibrium value of x under rational expectations. $\mathcal{X}_t(mc_t)$ *satisfies the following properties:*

1. If $\omega \in (0, 0.5]$, $\mathcal{X}_t$ is continuous;
2. $\mathcal{X}_t'(\mathrm{mc}_t) > 0$;
3. $|\mathcal{X}_t(\mathrm{mc}_t)| < |\mathcal{X}^*(\mathrm{mc}_t)|$;
4. As $|\mathrm{mc}_t| \to \infty$,
$$\left|\frac{\mathcal{X}_t(\mathrm{mc}_t)}{\mathrm{mc}_t}\right| \to \left|\frac{\mathcal{X}^*(\mathrm{mc}_t)}{\mathrm{mc}_t}\right| = \frac{1}{1-\omega}$$
5. If $\mathrm{mc}_t \in [-\ell_t, \ell_t]$, $\mathcal{X}_t'(\mathrm{mc}_t) < \mathcal{X}^{*\prime}(\mathrm{mc}_t)$;
6. If $|\mathrm{mc}_t| > \ell_t$, $\mathcal{X}_t'(\mathrm{mc}_t) > \mathcal{X}^{*\prime}(\mathrm{mc}_t)$.

Proof. See Appendix A.3. □

These properties are illustrated in Figure 1, which depicts the Phillips curve. The green dotted line corresponds to the Phillips curve under rationality. The blue line corresponds to the initial Phillips curve, under the assumption that the previous real marginal cost is located at the grey dotted line. mc_{-1} defines the boundary between the two regions in period 0.

As an example, we assume that the real marginal cost increases in period 0 to mc_0. As a result, the Phillips curve in period 1 features a wider Type-1 region, and the distance to the rational expectations Phillips curve decreases.

The model offers predictions on the second moments of the cross-sectional distribution of beliefs. Let $\Omega_t \equiv \mathrm{Var}_i[\mu_{i,t}]$, the dispersion of posterior beliefs on the passthrough ω. Using equation (10), it is easy to show that

$$\Omega_t = (1-\alpha_t)^2 \Omega_{t-1} + \alpha_t (1-\alpha_t) \frac{\overline{\sigma}^2}{x_{r,t-1}^2}.$$

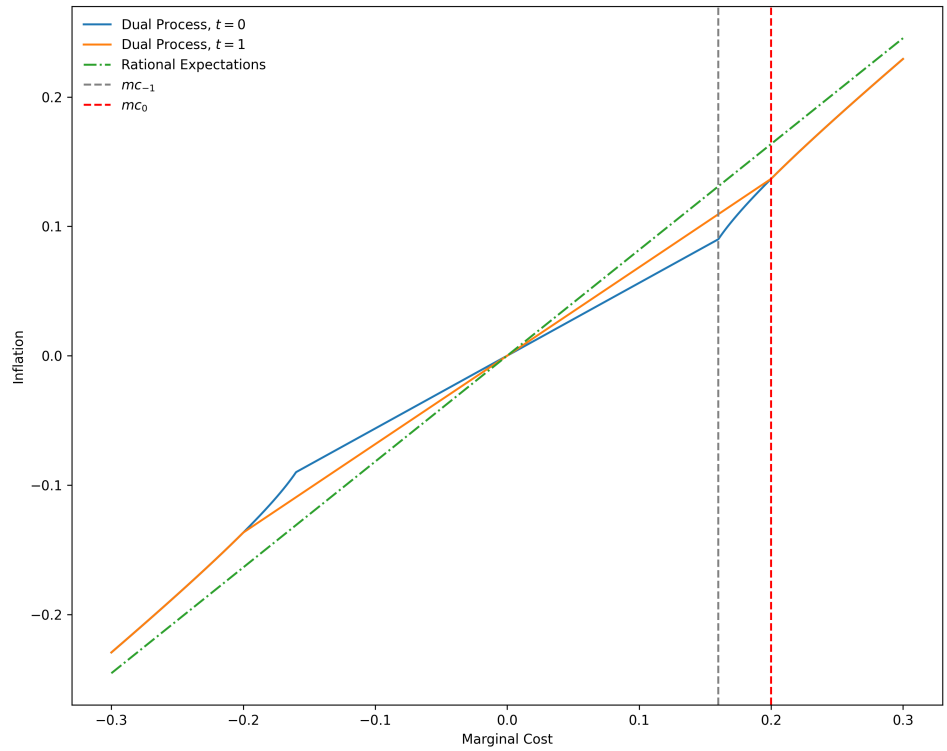

Figure 1: Phillips Curve

The dispersion of posterior beliefs on inflation is

$$\Delta_{e,t} = \Omega_t x_t^2 = \left(\frac{x_t^2}{x_{r,t}^2} \right) \Delta_{r,t},$$

where $\Delta_{r,t} \equiv \Omega_t x_{r,t}^2$. It follows that

$$\Delta_{e,t} = \begin{cases} \left(\frac{x_t^2}{x_{r,t-1}^2} \right) \Delta_{r,t-1}, & \text{if } x_t^2 \leq x_{r,t-1}^2 \\ \Delta_{r,t}, & \text{if } x_t^2 > x_{r,t-1}^2 \end{cases}, \tag{17}$$

and

$$\Delta_{r,t} = \begin{cases} \Delta_{r,t-1} & x_t^2 \leq x_{r,t-1}^2 \\ (1 - \alpha_t) \Delta_{r,t-1} + \alpha_t \overline{\sigma}^2 & x_t^2 > x_{r,t-1}^2 \end{cases}. \tag{18}$$

Equation (18) shows that reference disagreement, $\Delta_{r,t}$, is a non-decreasing sequence. Equation (17) then implies that belief dispersion co-moves with x_t. Since the Phillips curve predicts a positive equilibrium relation between x_t and π_t, belief dispersion also co-moves with π_t. This property is due to the fact that when inflation is high, forecasters rely on noisy private informa-

tion.

5 Optimal Policy

5.1 Planner's Problem

To solve the planner's problem, we first assume that

$$\mathrm{mc}_t = \lambda y_t + u_t, \tag{19}$$

where $\lambda \equiv \sigma + \psi$ is the slope of marginal cost in output (σ is the inverse elasticity of intertemporal substitution, ψ the inverse Frisch elasticity), y_t is the log output gap, and u_t is a markup shock that follows the AR(1) process

$$u_t = \rho u_{t-1} + \sigma_\nu \nu_t, \;\; \nu_t \sim \mathcal{N}(0,1).$$

We also consider the following period utility function

$$U_t = -\frac{\lambda}{2} y_t^2 - \frac{\gamma}{2}\left(\frac{1-\omega}{\omega}\right)\pi_t^2 - \left(\frac{\gamma\omega}{2}\right)\frac{x_t^2}{x_{r,t}^2}\Delta_{r,t} - \frac{\iota}{2}\ln\left(\frac{1}{1-\alpha_t}\right), \tag{20}$$

where $\gamma \equiv \theta / [1 - \beta(1-\omega)]$ is the price-dispersion weight (θ is the demand elasticity) and $\vartheta \equiv \omega(\theta - 1)$ governs the weight on forecast accuracy. Appendix C derives both (19) and (20) from a second-order approximation of the utility function in a standard New Keynesian environment. The first two terms represent the efficiency costs of deviating from the flexible-price equilibrium and of price dispersion. The third term captures the additional price dispersion caused by heterogeneous expectations. The last term consists of the cognitive costs of activating Type 2.

The problem of the central bank can be written recursively as follows. Let $\boldsymbol{\zeta} \equiv (u, \overline{\mu}, r, \Delta_r)$. The Bellman equations of the central bank are

$$V(\boldsymbol{\zeta}) = \max\{V_1(\boldsymbol{\zeta}); V_2(\boldsymbol{\zeta})\},$$

where

$$V_1(\boldsymbol{\zeta}) = \max_{|m| \le (1-\overline{\mu})\sqrt{r}} \left\{U_1(m, \boldsymbol{\zeta}) + \beta \mathbb{E}\left[V(\boldsymbol{\zeta}') \mid m, \boldsymbol{\zeta}\right]\right\},$$

$$V_2(\boldsymbol{\zeta}) = \max_{|m| > (1-\overline{\mu})\sqrt{r}} \left\{U_2(m, \boldsymbol{\zeta}) + \beta \mathbb{E}\left[V(\boldsymbol{\zeta}') \mid m, \boldsymbol{\zeta}\right]\right\},$$

$$U_1(m,\boldsymbol{\zeta}) \equiv -\frac{1}{2\lambda}(m-u)^2 - \frac{\gamma}{2}\left[(1-\omega)+\frac{\Delta_r}{r}\right]\omega\left(\frac{1}{1-\overline{\mu}}m\right)^2,$$

$$\begin{aligned} U_2(m,\boldsymbol{\zeta}) \equiv & -\frac{1}{2\lambda}(m-u)^2 - \frac{\gamma}{2}\omega(1-\omega)\mathcal{X}_2^2(m,\boldsymbol{\zeta}) \\ & -\frac{\gamma}{2}\omega\left\{\frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\Delta_r + \left[1-\frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\right]\overline{\sigma}^2\right\} - \frac{\overline{\sigma}^2\vartheta}{2}\ln\left[\frac{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}{r}\right], \end{aligned}$$

and

$$\mathcal{X}_2(m,\boldsymbol{\zeta}) = \operatorname{sign}(m)\frac{|m|+\sqrt{m^2-4(1-\omega)(\omega-\overline{\mu})r}}{2(1-\omega)}.$$

The laws of motion are

$$\overline{\mu}' = \begin{cases} \overline{\mu}, & \text{if } |m| \leq (1-\overline{\mu})\sqrt{r} \\ \frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\overline{\mu} + \left[1-\frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\right]\omega, & \text{if } |m| > (1-\overline{\mu})\sqrt{r} \end{cases},$$

$$r' = \begin{cases} r, & \text{if } |m| \leq (1-\overline{\mu})\sqrt{r} \\ \mathcal{X}_2^2(m,\boldsymbol{\zeta}), & \text{if } |m| > (1-\overline{\mu})\sqrt{r} \end{cases},$$

$$\Delta_r' = \begin{cases} \Delta_r, & \text{if } |m| \leq (1-\overline{\mu})\sqrt{r} \\ \frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\Delta_r + \left[1-\frac{r}{\mathcal{X}_2^2(m,\boldsymbol{\zeta})}\right]\overline{\sigma}^2, & \text{if } |m| > (1-\overline{\mu})\sqrt{r} \end{cases},$$

and

$$u' = \rho u + \sigma_v v', \;\; v' \sim \mathcal{N}(0,1).$$

Now note that conditional on Type 1, only the exogenous state u is updated. Therefore m does not influence $V(\boldsymbol{\zeta}')$ if $m \leq (1-\overline{\mu})\sqrt{r}$. It follows that at an interior solution in the Type 1 region,

$$m = \frac{u}{1+\lambda\gamma\left[(1-\omega)+\frac{\Delta_r}{r}\right]\left(\frac{\omega}{1-\overline{\mu}}\right)\times\left(\frac{1}{1-\overline{\mu}}\right)}.$$

On the other hand, under rational expectations the optimal value of m is

$$m^{\mathrm{RE}} = \frac{u}{1+\lambda\gamma\left(\frac{\omega}{1-\omega}\right)}.$$

Suppose the output-inflation tradeoff is the same as under rational expectations, *i.e.*, $\overline{\mu} = \omega$. Then, conditional on Type 1, $|m| < \left|m^{\mathrm{RE}}\right|$. The reason is that high inflation increases belief dispersion and therefore price dispersion.

The value function inherits two structural properties from the symmetry of the shock process and the compactness of the state space.

Lemma 2 (Evenness and symmetry)**.** *Assume that the planner's problem admits a bounded solution V, and let $\Gamma(\boldsymbol{\zeta})$ denote the feasible correspondence for m. Then:*

1. *The value function is even in u:*

$$V(-u, \overline{\mu}, r, \Delta_r) = V(u, \overline{\mu}, r, \Delta_r)\,.$$

2. *There exists an optimal policy $m^*(\boldsymbol{\zeta})$ satisfying $\operatorname{sign}\big(m^*(u, \overline{\mu}, r, \Delta_r)\big) = \operatorname{sign}(u)$ for $u \neq 0$, and $m^*(0, \overline{\mu}, r, \Delta_r) = 0$ can be selected whenever zero is optimal.*

3. *Defining $a \equiv |m| \geq 0$, there exists a reduced policy rule $a^*(|u|, \overline{\mu}, r, \Delta_r)$ such that $m^*(u, \overline{\mu}, r, \Delta_r) = \operatorname{sign}(u)\, a^*(|u|, \overline{\mu}, r, \Delta_r)$.*

Proof. See Appendix A.4. □

The sign-matching result is intuitive: every term in the period utility other than the stabilization loss $(m-u)^2$ depends on m only through $|m|$ or m^2, and the same is true for the transition laws. Conditional on the magnitude of m, choosing the sign to match u minimizes the stabilization loss without affecting any other component of the objective. Part (3) reduces the dimensionality of the numerical problem by restricting the search to non-negative magnitudes.

Lemma 3 (Continuity of the value function)**.** *Assume that the state space $\mathcal{S}$ is compact and that the feasible correspondence $\Gamma(\boldsymbol{\zeta})$ is nonempty and compact-valued for every $\boldsymbol{\zeta} \in \mathcal{S}$. Then the value function V is continuous on $\mathcal{S}$.*

Proof. See Appendix A.5. □

Remark 1. Continuity of V implies that the reduced objective over $a = |m|$ is continuous at the regime boundary $a = (1-\overline{\mu})\sqrt{r}$. The key step is that $\mathcal{X}_2^2(m, \boldsymbol{\zeta}) \to r$ as $|m| \downarrow (1-\overline{\mu})\sqrt{r}$, so the Type 2 transitions and period utility converge to their Type 1 counterparts at the boundary.

5.2 Calibration

The model is calibrated at a quarterly frequency. The discount factor is $\beta = 0.99$, and the Calvo parameter is $\omega = 0.25$ (average price duration of four quarters). The cost-push shock has persistence $\rho = 0.8$.

The structural parameters are $\sigma = 1$ (log utility), $\psi = 1$ (unit Frisch elasticity), and $\theta = 9$ (steady-state markup of 12.5 percent). The implied reduced-form coefficients are $\lambda = \sigma + \psi = 2$, $\gamma = \theta / [1 - \beta (1 - \omega)] \approx 34.95$, and $\vartheta = \omega (\theta - 1) = 2$.

We choose the innovation standard deviation σ_ν such that, under rational-expectations optimal policy, the 95th percentile of annualized inflation equals $\overline{\pi} = 10$ percentage points. Under rational expectations the optimal policy is $m^{\mathrm{RE}} = \varphi u$, where $\varphi \equiv \omega / (\lambda \gamma \omega + 1 - \omega) \approx 0.014$, so annualized inflation is $400 \varphi u_t$. Requiring $1.96 \times 400 \varphi \sigma_\nu / \sqrt{1 - \rho^2} = \overline{\pi}$ gives $\sigma_\nu \approx 0.558$.

The cognitive-cost parameter $\overline{\sigma}^2 = (0.015)^2 = 2.25 \times 10^{-4}$ governs the signal-to-noise ratio at the boundary between Type 1 and Type 2. Because the economy is non-ergodic — the record variable r is non-decreasing — standard ergodic calibration arguments do not apply. We instead discipline $\overline{\sigma}^2$ through the initial conditions and their empirical counterparts.

The initial conditions are $\overline{\mu}_{-1} = 0$ and $r_{-1} = (\pi_{\text{thresh}} / \omega)^2 = 0.0016$, where $\pi_{\text{thresh}} = 0.01$ corresponds to an annualized attention threshold of 4 percent, consistent with the estimate of Pfäuti (2024). Initial belief dispersion is $\Delta_{r,-1} = (0.005)^2 = 2.5 \times 10^{-5}$, calibrated to match a pre-pandemic cross-sectional standard deviation of inflation expectations of approximately 2 percent in annualized terms (Coibion and Gorodnichenko (2025)). These initial conditions imply a prior standard deviation over the passthrough of $\sigma_{-1} = \sqrt{\overline{\sigma}^2 / r_{-1}} \approx 0.375$ and an initial cross-sectional variance of beliefs $\Omega_{-1} = \Delta_{r,-1} / r_{-1} \approx 0.016$.

5.3 Numerical Results

Figure 2 plots the optimal policy function $m^*(u)$ at the initial state $(\overline{\mu}_{-1}, r_{-1}, \Delta_{r,-1})$ for three planners: the rational-expectations (RE) benchmark, a naive central bank, and the baseline planner. Under RE the policy is linear, $m^{\mathrm{RE}} = \varphi u$. Both non-RE planners share a qualitative feature absent from the RE solution: a flat region around $u = 0$ in which optimal marginal cost does not respond to the shock. In that region the constraint $|m| \leq (1 - \overline{\mu}) \sqrt{r}$ binds and the economy remains in Type 1 — the central bank accommodates the shock just enough to avoid triggering learning, keeping beliefs and the record unchanged.

The naive planner's period utility is

$$\mathcal{U}_t^{\mathrm{Naive}} (\mathrm{mc}_t) = -\frac{1}{2\lambda} (\mathrm{mc}_t - u_t)^2 - \frac{\gamma}{2} \omega (1 - \omega) [\mathcal{X}_t (\mathrm{mc}_t)]^2 .$$

The naive planner correctly recognizes the shape of the Phillips curve but does not internalize the welfare effects of belief dispersion and cognitive costs. Once the shock is large enough to push the economy into Type 2, the naive and baseline policies diverge: the baseline planner chooses a lower $|m^*|$ because higher inflation triggers heterogeneous belief updating (Δ_r rises)

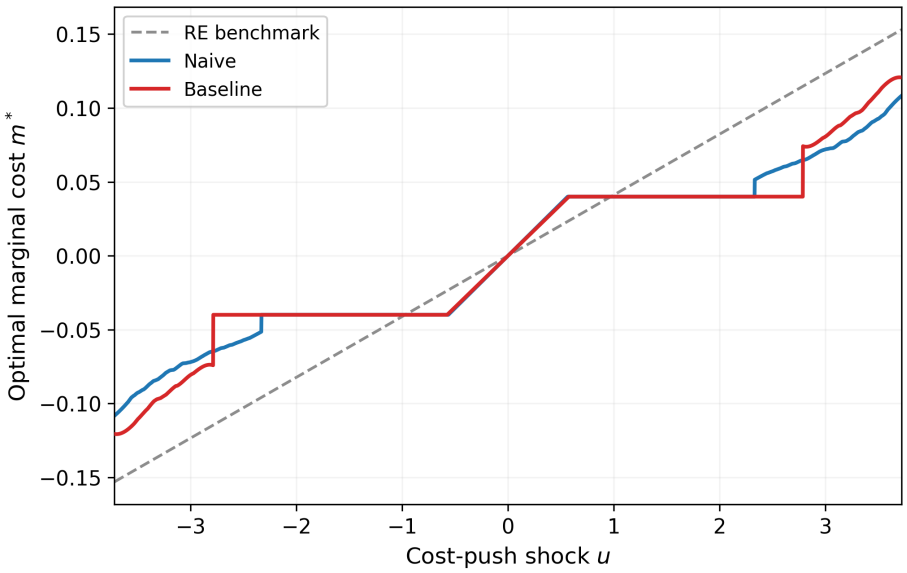

Figure 2: Optimal policy function $m^*(u)$ at the initial state. The dashed line is the RE benchmark. The flat region around the origin corresponds to Type 1 (no learning). Outside that region, the baseline planner (red) chooses lower $|m^*|$ than the naive planner (blue) because it internalizes the costs of belief dispersion and cognitive effort.

and imposes cognitive costs on firms.

Figure 3 reports generalized impulse responses under the baseline planner for a small and a large cost-push shock. In both cases, belief dispersion $\Delta_{e,t}$ rises on impact because inflation raises x_t^2 and $\Delta_{e,t} = (x_t^2/x_{r,t}^2)\Delta_{r,t}$, even when the reference disagreement $\Delta_{r,t}$ itself is unchanged.

Under the small shock, the optimal policy keeps $|m^*|$ below the regime boundary: the economy remains in Type 1 and no learning occurs at impact. The average prior mean $\overline{\mu}_t$ is initially unchanged, but rises gradually as the persistence of u causes some simulated paths to eventually cross the regime boundary and trigger Type 2 updating. The GIRF, which averages across paths, registers this delayed learning as a slow drift in $\overline{\mu}_t$.

Under the large shock, the optimal policy pushes $|m^*|$ past the regime boundary at impact, triggering Type 2 immediately. The average prior mean jumps to approximately $\overline{\mu} \approx 0.22$, close to the Calvo parameter $\omega = 0.25$ that governs the RE Phillips curve slope. Because the slope is now near its RE value, the optimal inflation path converges almost immediately to the RE benchmark — the two paths are nearly indistinguishable after the first quarter.

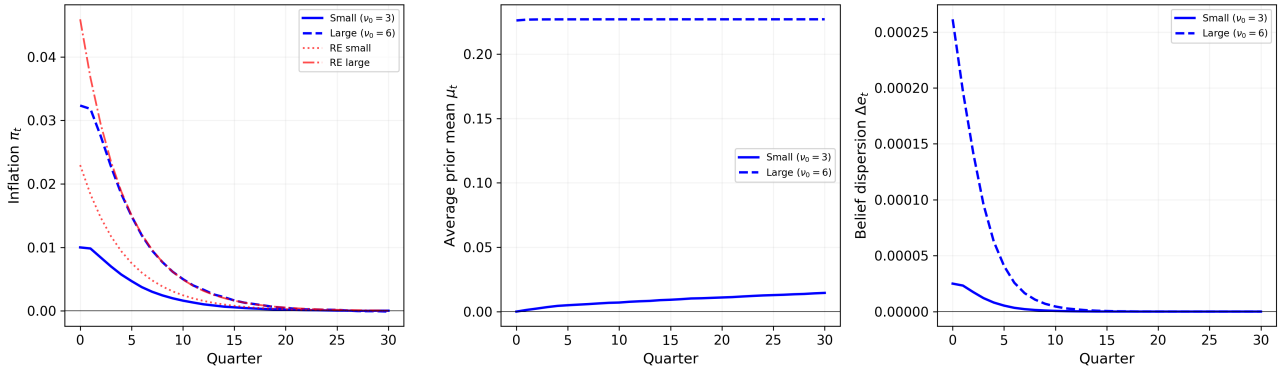

Figure 3: Generalized impulse responses under the baseline planner. The small shock ($\nu_0 = 3$) keeps the economy in Type 1 at impact; the large shock ($\nu_0 = 6$) triggers Type 2. Belief dispersion $\Delta_{e,t}$ rises on impact in both cases because higher inflation amplifies cross-sectional disagreement. Under the large shock, $\overline{\mu}$ jumps close to ω and the inflation path converges to the RE benchmark.

6 Conclusion

We develop a model where firms form inflation expectations through dual-process reasoning. When the economic state is familiar, they rely on a low-cost heuristic (Type 1) that attenuates the response of inflation to marginal costs. When the state becomes unfamiliar, firms pay the cognitive cost of acquiring private signals (Type 2), and the Phillips curve steepens toward its rational-expectations counterpart. The boundary between regimes is endogenous and history-dependent.

The model generates three empirical patterns: state-dependent attention to inflation, cross-sectional belief dispersion that co-moves with inflation, and a nonlinear Phillips curve whose nonlinearity operates entirely through expectations. Inflation carries welfare costs absent from the canonical New Keynesian model — the cognitive cost of triggering Type 2 and the efficiency loss from heterogeneous beliefs — and both costs push optimal inflation below the rational-expectations benchmark.

Two extensions are natural. The first is to consider forward-looking firms. The second is an empirical investigation of firm-level inflation expectations. The model's central predictions — that attention, forecast accuracy, and belief dispersion are jointly state-dependent — are testable with firm survey data. Matching the model to such data would discipline the cognitive-cost parameters and sharpen the quantitative implications for optimal policy.

A Proofs

A.1 Proof of Lemma 1

Proof. Using Bayes' updating formulas,

$$\mu_{i,t} = \mu_{i,t-1} + \frac{\sigma_{i,t-1}^2 x_t}{\sigma_{i,t-1}^2 x_t^2 + \sigma_{\epsilon,i,t}^2} \left(\omega x_t + \sigma_{\epsilon,i,t}\epsilon_{i,t} - \mu_{i,t-1} x_t\right),$$

and

$$\sigma_{i,t}^2 = \min\left\{\frac{\overline{\sigma}^2}{x_t^2}; \sigma_{i,t-1}^2\right\}.$$

Define

$$\alpha_{i,t} \equiv 1 - \frac{\sigma_{i,t}^2}{\sigma_{i,t-1}^2}.$$

Then

$$\alpha_{i,t} = \max\left\{0; 1 - \frac{\overline{\sigma}^2}{\sigma_{i,t-1}^2 x_t^2}\right\}.$$

Using $\sigma_{i,t-1}^2 = \overline{\sigma}^2 / x_{i,r,t-1}^2$, this becomes

$$\alpha_{i,t} = \max\left\{0; 1 - \frac{x_{i,r,t-1}^2}{x_t^2}\right\}.$$

Also, with the same substitution,

$$\sqrt{\alpha_{i,t}\left(1-\alpha_{i,t}\right)}\sigma_{i,t-1} = \sqrt{\frac{\alpha_{i,t}\left(1-\alpha_{i,t}\right)}{x_{i,r,t-1}^2}}\,\overline{\sigma}.$$

Substituting into the posterior-mean equation yields

$$\mu_{i,t} = \mu_{i,t-1} + \alpha_{i,t}\left(\omega - \mu_{i,t-1}\right) + \operatorname{sign}\left(x_t\right)\sqrt{\frac{\alpha_{i,t}\left(1-\alpha_{i,t}\right)}{x_{i,r,t-1}^2}}\overline{\sigma}\epsilon_{i,t}.$$

Finally,

$$\sigma_{i,t}^2 = \min\left\{\frac{\overline{\sigma}^2}{x_t^2}; \frac{\overline{\sigma}^2}{x_{i,r,t-1}^2}\right\} = \frac{\overline{\sigma}^2}{x_{i,r,t}^2},$$

with

$$x_{i,r,t} = \begin{cases} x_t, & \text{if } x_t^2 > x_{i,r,t-1}^2 \\ x_{i,r,t-1}, & \text{if } x_t^2 \leq x_{i,r,t-1}^2 \end{cases}.$$

□

A.2 Proof of Theorem 1

Proof. Equilibrium expectations are

$$\begin{aligned} &\overline{\mathbb{E}}_t \pi_t = \overline{\mu}_t \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right) \\ \iff &\overline{\mathbb{E}}_t \pi_t = \begin{cases} \overline{\mu}_{t-1} \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right), & \text{if } \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right)^2 \leq x_{r,t-1}^2 \\ \left(\overline{\mu}_{t-1} + \left(1 - \frac{x_{r,t-1}^2}{\left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right)^2}\right)\left(\omega - \overline{\mu}_{t-1}\right)\right)\left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right) & \text{if } \left(\mathrm{mc}_t + \overline{\mathbb{E}}_t \pi_t\right)^2 > x_{r,t-1}^2 \end{cases} \end{aligned}$$

Consider the second branch. We have

$$\begin{aligned} x_t - \mathrm{mc}_t &= \left[\overline{\mu}_{t-1} + \left(1 - \frac{x_{r,t-1}^2}{x_t^2}\right)\left(\omega - \overline{\mu}_{t-1}\right)\right] x_t \\ \iff \left(1 - \overline{\mu}_{t-1}\right) x_t^3 &= \mathrm{mc}_t x_t^2 + \left(x_t^2 - x_{r,t-1}^2\right)\left(\omega - \overline{\mu}_{t-1}\right) x_t. \end{aligned}$$

Evidently the zero solution is not relevant. So the equation is

$$\begin{aligned} &\left(1 - \overline{\mu}_{t-1}\right) x_t^2 = \mathrm{mc}_t x_t + \left(\omega - \overline{\mu}_{t-1}\right) x_t^2 - \left(\omega - \overline{\mu}_{t-1}\right) x_{r,t-1}^2 \\ \iff &\left(1 - \omega\right) x_t^2 - \mathrm{mc}_t x_t + \left(\omega - \overline{\mu}_{t-1}\right) x_{r,t-1}^2 = 0 \\ \iff &x_t = \frac{\mathrm{mc}_t \pm \sqrt{\mathrm{mc}_t^2 - 4\left(1 - \omega\right)\left(\omega - \overline{\mu}_{t-1}\right) x_{r,t-1}^2}}{2\left(1 - \omega\right)}. \end{aligned}$$

For a solution in this branch to exist, we need

$$\mathrm{mc}_t^2 \geq 4\left(1 - \omega\right)\left(\omega - \overline{\mu}_{t-1}\right) x_{r,t-1}^2.$$

The solution in the first branch is

$$\overline{\mathbb{E}}_t \pi_t = \frac{\overline{\mu}_{t-1}}{1 - \overline{\mu}_{t-1}} \mathrm{mc}_t.$$

Therefore, in the first branch,

$$x_t = \frac{1}{1-\overline{\mu}_{t-1}}\mathrm{mc}_t,$$

from which it follows that a solution on the first branch exists if

$$\mathrm{mc}_t^2 \leq \left(1-\overline{\mu}_{t-1}\right)^2 x_{r,t-1}^2.$$

Now consider the second branch.

$$x_t = \frac{\mathrm{mc}_t \pm \sqrt{\mathrm{mc}_t^2 - 4\left(1-\omega\right)\left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2}}{2\left(1-\omega\right)}.$$

Now

$$\left(1-\overline{\mu}_{t-1}\right)^2 - 4\left(1-\omega\right)\left(\omega-\overline{\mu}_{t-1}\right) = \left(1+\overline{\mu}_{t-1}-2\omega\right)^2 \geq 0.$$

Now consider the quadratic

$$Q\left(x\right) = \left(1-\omega\right)x^2 - \mathrm{mc}_t x + \left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2.$$

The smallest root is valid iff

$$\begin{aligned}
& Q\left(-\left|x_{r,t-1}\right|\right) < 0 \\
\iff & \left(1-\omega\right)x_{r,t-1}^2 + \mathrm{mc}_t\left|x_{r,t-1}\right| + \left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2 < 0 \\
\iff & \mathrm{mc}_t < -\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right|
\end{aligned}$$

and the largest root is valid iff

$$\begin{aligned}
& Q\left(\left|x_{r,t-1}\right|\right) < 0 \\
\iff & \left(1-\omega\right)x_{r,t-1}^2 - \mathrm{mc}_t\left|x_{r,t-1}\right| + \left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2 < 0 \\
\iff & \left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| < \mathrm{mc}_t
\end{aligned}$$

Therefore, for the solution to be admissible we need $\mathrm{mc}_t^2 > \left(1-\overline{\mu}_{t-1}\right)^2 x_{r,t-1}^2$, which means that

the solution is unique (because $\left(1-\overline{\mu}_{t-1}\right)^2 > 4\left(1-\omega\right)\left(\omega-\overline{\mu}_{t-1}\right)$).

$$\mathcal{X}_t\left(\mathrm{mc}_t\right) = \begin{cases} \frac{\mathrm{mc}_t - \sqrt{\mathrm{mc}_t^2 - 4(1-\omega)\left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2}}{2(1-\omega)} & \mathrm{mc}_t < -\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| \\ \frac{1}{1-\overline{\mu}_{t-1}}\mathrm{mc}_t & -\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| \leq \mathrm{mc}_t \leq \left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| \\ \frac{\mathrm{mc}_t + \sqrt{\mathrm{mc}_t^2 - 4(1-\omega)\left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2}}{2(1-\omega)} & \mathrm{mc}_t \geq \left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| \end{cases}$$

Therefore the Phillips curve is

$$\mathcal{PC}_t\left(\mathrm{mc}_t\right) = \omega\mathcal{X}_t\left(\mathrm{mc}_t\right)$$

□

A.3 Proof of Theorem 2

Proof. For 1, Now note that at the thresholds in the no-learning region, $\mathcal{X}_t\left(\mathrm{mc}_t\right) = \pm\left|x_{r,t-1}\right|$.

$$\mathcal{X}_t\left(\mathrm{mc}_t\right) = x_{r,t-1}$$

In the limit of the learning regions, the discriminant is

$$\begin{aligned} \Delta &= \left(1-\overline{\mu}_{t-1}\right)^2 x_{r,t-1}^2 - 4\left(1-\omega\right)\left(\omega-\overline{\mu}_{t-1}\right)x_{r,t-1}^2 \\ &= \left(1+\overline{\mu}_{t-1}-2\omega\right)^2 x_{r,t-1}^2. \end{aligned}$$

Suppose $1+\overline{\mu}_{t-1}-2\omega > 0$. Then $\sqrt{\Delta} = \left(1+\overline{\mu}_{t-1}-2\omega\right)\left|x_{r,t-1}\right|$, and

$$\begin{aligned} \mathcal{X}_t\left(-\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right|\right) &= \frac{-\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right| - \left(1+\overline{\mu}_{t-1}-2\omega\right)\left|x_{r,t-1}\right|}{2\left(1-\omega\right)} \\ &= -\left|x_{r,t-1}\right|. \end{aligned}$$

Same for the other side. Suppose instead that $1+\overline{\mu}_{t-1}-2\omega < 0$. Then

$$\sqrt{\Delta} = -\left(1+\overline{\mu}_{t-1}-2\omega\right)\left|x_{r,t-1}\right|$$

and

$$\mathcal{X}_t\left(-\left(1-\overline{\mu}_{t-1}\right)\left|x_{r,t-1}\right|\right) = \frac{\overline{\mu}_{t-1}-\omega}{1-\omega}\left|x_{r,t-1}\right|,$$

which means that the Phillips curve is not continuous. This happens because of too many complementarities. When $\omega \leq 0.5$, $1+\overline{\mu}_{t-1}-2\omega \geq 0$ for all $\overline{\mu}_{t-1}$.

For 2, 5, and 6,

$$\mathcal{X}_t'\left(\mathrm{mc}_t\right) = \begin{cases} \frac{1}{2(1-\omega)}\left(1+\frac{|\mathrm{mc}_t|}{\sqrt{\mathrm{mc}_t^2-\Delta_t}}\right), & |\mathrm{mc}_t| > \ell_t \\ \frac{1}{1-\overline{\mu}_{t-1}} & |\mathrm{mc}_t| \leq \ell_t \end{cases}.$$

First, in the no-learning region,

$$\frac{1}{1-\overline{\mu}_{t-1}} < \frac{1}{1-\omega} \iff 1-\omega < 1-\overline{\mu}_{t-1} \iff \overline{\mu}_{t-1} < \omega.$$

In the learning regions,

$$\begin{aligned} & \frac{1}{2\left(1-\omega\right)}\left(1+\frac{|\mathrm{mc}_t|}{\sqrt{\mathrm{mc}_t^2-\Delta_t}}\right) > \frac{1}{1-\omega} \\ \iff & \frac{1}{2}\left(1+\frac{|\mathrm{mc}_t|}{\sqrt{\mathrm{mc}_t^2-\Delta_t}}\right) > 1 \\ \iff & |\mathrm{mc}_t| > \sqrt{\mathrm{mc}_t^2-\Delta_t} \\ \iff & \mathrm{mc}_t^2 > \mathrm{mc}_t^2 - \Delta_t \\ \iff & 0 > -\Delta_t. \end{aligned}$$

For 3 and 4,

$$\left|\mathcal{X}_t\left(\mathrm{mc}_t\right)\right| = \begin{cases} \frac{|\mathrm{mc}_t|+\sqrt{\mathrm{mc}_t^2-\Delta_t}}{2(1-\omega)}, & \mathrm{mc}_t < -\ell_t \\ \frac{1}{1-\overline{\mu}_{t-1}}\left|\mathrm{mc}_t\right|, & -\ell_t \leq \mathrm{mc}_t \leq \ell_t \\ \frac{|\mathrm{mc}_t|+\sqrt{\mathrm{mc}_t^2-\Delta_t}}{2(1-\omega)}, & \mathrm{mc}_t > \ell_t \end{cases}$$

and

$$\begin{aligned} & \frac{|\mathrm{mc}_t|+\sqrt{\mathrm{mc}_t^2-\Delta_t}}{2\left(1-\omega\right)} < \frac{1}{1-\omega}\left|\mathrm{mc}_t\right| \\ \iff & +\sqrt{\mathrm{mc}_t^2-\Delta_t} < |\mathrm{mc}_t| \\ \iff & \mathrm{mc}_t^2 - \Delta_t < \mathrm{mc}_t^2 \\ \iff & -\Delta_t < 0, \end{aligned}$$

which is true. Moreover, in the learning region,

$$\left|\frac{\mathcal{X}_t\left(\mathrm{mc}_t\right)}{\mathrm{mc}_t}\right| = \frac{1+\sqrt{1-\frac{\Delta_t}{\mathrm{mc}_t^2}}}{2\left(1-\omega\right)} \to \frac{1}{1-\omega}$$

as $|\mathrm{mc}_t| \to \infty$. □

A.4 Proof of Lemma 2

Proof. Fix a state $\boldsymbol{\zeta} = (u, \overline{\mu}, r, \Delta_r)$ and a feasible magnitude $a \geq 0$. Consider the two feasible controls $m = +a$ and $m = -a$. The feasibility constraints depend only on $|m|$. Moreover, every term in U_1 and U_2 other than $(m-u)^2$ depends on m only through $|m|$ or m^2. The same is true for the transition laws of $(\overline{\mu}', r', \Delta_r')$, because they depend on m only through $\mathcal{X}_2^2(m, \boldsymbol{\zeta})$, and $\mathcal{X}_2^2$ depends only on $|m|$.

Therefore, conditional on the magnitude a, the only difference between choosing $m = +a$ and choosing $m = -a$ is the stabilization term. If $u > 0$,

$$(a-u)^2 \leq (a+u)^2,$$

with strict inequality whenever $a > 0$. Hence $m = +a$ weakly dominates $m = -a$. If $u < 0$, then

$$(-a-u)^2 = (a-|u|)^2 \leq (a+|u|)^2 = (a-u)^2,$$

so $m = -a$ weakly dominates $m = +a$. Thus, for any feasible magnitude a, one can select the sign of m to match the sign of u without lowering the objective. This proves part (2).

Substituting the optimal sign choice $m = \mathrm{sign}(u)\, a$ yields $(m-u)^2 = (a-|u|)^2$, while all other terms continue to depend only on a. Hence the reduced problem depends on the exogenous state only through $|u|$, and the induced optimal magnitude can be written as $a^*(|u|, \overline{\mu}, r, \Delta_r)$. This proves part (3).

It remains to show that V is even in u. Let T denote the Bellman operator. Suppose W is any bounded function that is even in u. By the argument above, after choosing the optimal sign of m, the current-period objective depends on the state only through $|u|$, and the endogenous next state $(\overline{\mu}', r', \Delta_r')$ depends only on the chosen magnitude a. The only remaining dependence on the sign of u is through

$$u' \mapsto W\left(\rho u + \sigma_\nu \nu', \overline{\mu}', r', \Delta_r'\right).$$

Because W is even in its first argument and ν' is symmetrically distributed,

$$\mathbb{E}\left[W\left(\rho u + \sigma_\nu \nu', \overline{\mu}', r', \Delta_r'\right)\right] = \mathbb{E}\left[W\left(-\rho u + \sigma_\nu \nu', \overline{\mu}', r', \Delta_r'\right)\right].$$

Therefore $(TW)(-u, \overline{\mu}, r, \Delta_r) = (TW)(u, \overline{\mu}, r, \Delta_r)$: the Bellman operator maps even functions into even functions. Starting value-function iteration from any even initial guess (for instance, the zero function), every iterate is even, and so is the fixed point V. This proves part (1). □

A.5 Proof of Lemma 3

Proof. Let $C(\mathcal{S})$ denote the space of continuous bounded functions on $\mathcal{S}$, endowed with the sup norm. We show that the Bellman operator maps $C(\mathcal{S})$ into itself. Since discounting implies a contraction on bounded functions, the unique fixed point then belongs to $C(\mathcal{S})$.

Fix any $W \in C(\mathcal{S})$. Consider first the Type 1 branch. For $|m| \leq (1-\overline{\mu})\sqrt{r}$, the one-period payoff $U_1(m, \boldsymbol{\zeta})$ is continuous in $(m, \boldsymbol{\zeta})$, and the next-state map is continuous:

$$\boldsymbol{\zeta}_1' = \left(\rho u + \sigma_\nu \nu', \overline{\mu}, r, \Delta_r\right).$$

Because W is continuous, the continuation term

$$(m, \boldsymbol{\zeta}) \mapsto \beta \mathbb{E}\left[W\left(\boldsymbol{\zeta}_1'\right) \mid m, \boldsymbol{\zeta}\right]$$

is continuous. Hence the Type 1 objective is continuous in $(m, \boldsymbol{\zeta})$.

Now consider the Type 2 branch. On the open region $|m| > (1-\overline{\mu})\sqrt{r}$, the function $\mathcal{X}_2(m, \boldsymbol{\zeta})$ is well defined and continuous, so $U_2(m, \boldsymbol{\zeta})$ is continuous there. The Type 2 transition laws are also continuous on that region because they are continuous functions of $\mathcal{X}_2^2(m, \boldsymbol{\zeta})$.

It remains to verify continuity at the boundary $|m| = (1-\overline{\mu})\sqrt{r}$. Write $\bar{a} \equiv (1-\overline{\mu})\sqrt{r}$. At $|m| = \bar{a}$,

$$\begin{aligned}\bar{a}^2 - 4(1-\omega)(\omega - \overline{\mu})r &= \left[(1-\overline{\mu})^2 - 4(1-\omega)(\omega-\overline{\mu})\right] r \\ &= (1+\overline{\mu}-2\omega)^2 r,\end{aligned}$$

so

$$\mathcal{X}_2^2(m, \boldsymbol{\zeta}) = r \qquad \text{whenever } |m| = \bar{a}.$$

Therefore, as $|m| \downarrow \bar{a}$ from the Type 2 region,

$$\overline{\mu}' \to \overline{\mu}, \qquad r' \to r, \qquad \Delta_r' \to \Delta_r,$$

which coincides exactly with the Type 1 transition law at the boundary. Period utility is continuous at the boundary by direct verification, and since W is continuous, the continuation term is also continuous there. It follows that the full-branch objective is continuous across the Type 1/Type 2 boundary.

Thus, for every continuous W, both branch objectives are jointly continuous in $(m, \boldsymbol{\zeta})$. Since the feasible correspondence is compact-valued and continuous by construction, Berge's maximum theorem implies that the value generated by each branch is continuous in $\boldsymbol{\zeta}$. Taking the maximum of two continuous functions yields another continuous function, so $TW \in C(\mathcal{S})$.

Finally, the Bellman operator is a contraction with modulus $\beta \in (0, 1)$ on bounded functions. Hence it has a unique fixed point. Because T preserves continuity, the fixed point V lies in $C(\mathcal{S})$. □

B Computational Appendix

B.1 Solution Method

The planner's problem is solved by value function iteration on the state space $\boldsymbol{\zeta} = (u, \overline{\mu}, r, \Delta_r)$. The value function is approximated on a rectangular grid with multilinear interpolation. We use multilinear rather than Chebyshev interpolation for two reasons. First, the value function inherits kinks from the regime boundary $|m| = (1 - \overline{\mu})\sqrt{r}$; polynomial approximations produce Gibbs-type oscillations near kinks that contaminate the global fit. Second, tensor-product Chebyshev grids scale poorly in four dimensions.

The cost-push shock u is discretized on a uniform grid of 41 points over $\left[-4\sigma_\nu / \sqrt{1-\rho^2},\, 4\sigma_\nu / \sqrt{1-\rho^2}\right]$ $[-0.41,\, 0.41]$. The average prior mean $\overline{\mu}$ is placed on a uniform grid of 16 points over $[0, \omega]$. The record variable r is placed on a logarithmically spaced grid of 16 points over $[r_{-1}, r_{\max}]$, where $r_{\max}$ is initially set to $10r_{-1}$ and widened if the optimal policy pushes r near the boundary. Belief dispersion Δ_r is placed on a uniform grid of 11 points over $\left[0, \overline{\sigma}^2\right]$. The baseline problem thus involves $41 \times 16 \times 16 \times 11 = 115{,}456$ state points. The naive problem (three states: u, $\overline{\mu}$, r) and the rational prior problem (three states: u, r, Δ_r; $\overline{\mu} = \omega$) use finer grids of 51, 21, and 21 (or 16) points per dimension.

B.2 Optimization and Expectations

The policy variable m is optimized by grid search over 401 uniformly spaced points on $[-u_{\max}, u_{\max}]$, where $u_{\max} = 4\sigma_\nu / \sqrt{1-\rho^2}$. Grid search is preferred to gradient-based methods because the regime structure renders the objective non-concave in m: the period utility switches between U_1 and U_2 at the regime boundary, and the continuation value inherits the non-concavity. For each state point, the objective $U_j(m, \boldsymbol{\zeta}) + \beta \mathbb{E}\left[V(\boldsymbol{\zeta}') \mid m, \boldsymbol{\zeta}\right]$ is evaluated at all grid points in the respective regime ($j = 1$ for $|m| \leq (1 - \overline{\mu})\sqrt{r}$, $j = 2$ otherwise), and the overall maximum is selected.

The conditional expectation $\mathbb{E}\left[V(\boldsymbol{\zeta}') \mid m, \boldsymbol{\zeta}\right]$ is computed by Gauss-Hermite quadrature with 9 nodes, using the probabilist's convention (nodes and weights for $\int f(z)\,\phi(z)\,dz$, where ϕ is the standard normal density). Since the endogenous states $\overline{\mu}'$, r', and Δ_r' are deterministic conditional on $(m, \boldsymbol{\zeta})$, the only source of randomness in the transition is the innovation ν', which enters through $u' = \rho u + \sigma_\nu \nu'$. Each quadrature node produces a value of u' at which the continuation value is interpolated using the multilinear scheme. Quadrature nodes that map u' outside the grid are clamped to the boundary. Sensitivity checks confirm that widening the grid from ± 4 to ± 5 standard deviations does not affect the interior policy function.

B.3 Convergence

The iteration terminates when the sup-norm of the value function update falls below 10^{-6}. Convergence is accelerated by Howard's policy iteration: between policy updates, the current policy is evaluated for 25 steps. The policy function $m^*(\zeta)$ is also monitored to ensure that it stabilizes, as the value function can converge while the policy oscillates in flat regions.

The solution is verified against three diagnostics. First, in the rational prior problem ($\overline{\mu} = \omega$), the computed policy converges to $m^* = u/(1 + \lambda\gamma\omega/(1-\omega))$ as $\overline{\sigma}^2 \to 0$, recovering the rational-expectations benchmark. Second, period utility is verified to be continuous at the regime boundary $|m| = (1-\overline{\mu})\sqrt{r}$ for a sample of state points. Third, the laws of motion from both branches agree at the boundary: $\overline{\mu}' = \overline{\mu}$, $r' = r$, and $\Delta_r' = \Delta_r$.

B.4 Generalized Impulse Response Functions

Because the economy is non-ergodic, we report generalized impulse response functions (GIRFs) following the method of Koop et al. (1996). The initial state is fixed at $\left(\overline{\mu}_{-1}, r_{-1}, \Delta_{r,-1}\right)$ from the calibration, with $u_{-1} = 0$. We draw $N = 10,000$ paths of future innovations $\{\nu_1', \ldots, \nu_T'\}$, shared across all scenarios. For each draw, the economy is simulated forward under two scenarios: a baseline with no initial shock ($\nu_0 = 0$) and a shocked path ($\nu_0 = \nu_{\text{large}}$, calibrated so that the optimal policy at impact pushes $|m^*|$ past the regime boundary). From $t = 1$ onward, both scenarios use the same innovation draws. The GIRF for each variable is the cross-simulation average of the difference between the shocked and baseline paths. Common random numbers across scenarios reduce the Monte Carlo variance of the difference. We simulate over a horizon of $T = 60$ quarters.

C Microfoundation

This appendix derives the reduced-form welfare function (20) and the forecaster's payoff from a standard New Keynesian environment with CES demand and nominal rigidities.

C.1 Household and Demand

A representative household has period utility

$$\mathcal{U}_t = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\psi}}{1+\psi} - \int_0^1 \mathcal{I}_{i,t}\, di,$$

where $\sigma > 0$ is the inverse elasticity of intertemporal substitution, $\psi > 0$ is the inverse Frisch elasticity, $\mathcal{I}_{i,t}$ is the cognitive cost of forecaster i, and

$$C_t = \left(\int_0^1 C_{i,t}^{\frac{\theta-1}{\theta}} \, di \right)^{\frac{\theta}{\theta-1}}, \qquad \theta > 1,$$

is the CES aggregate with demand elasticity θ. Demand for variety i is $C_{i,t} = (P_{i,t}/P_t)^{-\theta} C_t$.

The household supplies labor to a competitive labor market and owns all firms. The Lagrangian in period t is

$$\mathcal{L}_t = \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\psi}}{1+\psi} - \int_0^1 \mathcal{I}_{i,t} \, di + \Lambda_t \left(W_t N_t + \int_0^1 \Pi_{i,t} \, di + R_{t-1} B_{t-1} - P_t C_t - B_t \right) \right],$$

where Λ_t is the multiplier on the budget constraint.

C.2 Firm Profits and the Forecaster's Problem

Each forecaster chooses expectations before firms set prices. The utility of forecaster i is $\Lambda_t \Pi_{i,t} - \mathcal{I}_{i,t}$. The profit of firm i, valued at the stochastic discount factor, is

$$\Lambda_t \Pi_{i,t} = C_t^{1-\sigma} \left(\frac{P_{i,t}}{P_t} \right)^{-\theta} \left(\frac{P_{i,t}}{P_t} - \frac{\theta-1}{\theta} e^{m_t} \right),$$

where $m_t \equiv \xi_t + w_t - p_t$ is the log deviation of real marginal cost from its steady-state value, ξ_t is the log deviation of the marginal product markup, and $w_t - p_t$ is the log real wage. At the zero-inflation steady state, the markup is $\theta/(\theta-1)$, the real wage is one, and $P_{i,t}/P_t = 1$.

Define $q_{i,t} \equiv p_{i,t} - p_t$. A second-order expansion around the zero-inflation steady state yields

$$\begin{aligned} \frac{\Lambda_t \Pi_{i,t}}{\overline{C}^{1-\sigma}} \approx \frac{1}{\theta} + \frac{1-\sigma}{\theta} c_t - \frac{\theta-1}{\theta} m_t + \frac{(1-\sigma)^2}{2\theta} c_t^2 \\ - \frac{\theta-1}{\theta}(1-\sigma) c_t m_t + \frac{\theta-1}{2\theta} m_t^2 - \frac{\theta-1}{2} \left(q_{i,t} - m_t \right)^2. \end{aligned}$$

The last term is the only one that depends on the individual price $q_{i,t}$. Normalizing $\overline{C}^{1-\sigma} = 1$, the ex-post utility of each forecaster is

$$\hat{v}_{i,t} = -\frac{\theta-1}{2} \left[p_{i,t} - (p_t + m_t) \right]^2 - \mathcal{I}_{i,t}.$$

Substituting the pricing rule (2) and the Calvo structure (3), the term $p_{i,t-1} - p_{t-1} - (\pi_t + m_t)$ for non-resetting firms does not depend on the forecast. The forecast-dependent component of the resetting firm's payoff reduces to

$$-\frac{\omega(\theta-1)}{2}\left(\mathbb{E}_{i,t}\pi_t - \pi_t\right)^2 - \mathcal{I}_{i,t},$$

yielding the relation

$$\vartheta = \omega\,(\theta - 1) \tag{21}$$

used in the firm's problem (6).

C.3 Second-Order Welfare Approximation

From labor market clearing, $N_t = Y_t \Delta_{p,t}$, where $\Delta_{p,t} = \int_0^1 (P_{i,t}/P_t)^{-\theta}\, di$ is the price dispersion index. A second-order expansion of household utility gives

$$\mathcal{U}_t - \overline{\mathcal{U}} \approx -\frac{\sigma+\psi}{2}\,\hat{y}_t^2 - \frac{\theta}{2}\,\mathrm{Var}_i\left[p_{i,t}\right] + \text{t.i.p.},$$

where $\hat{y}_t$ is the log output gap and t.i.p. collects terms independent of policy. Cross-sectional price dispersion decomposes as

$$\mathrm{Var}_i\left[p_{i,t}\right] = \omega\,\Delta_{e,t} + \frac{1-\omega}{\omega}\,\pi_t^2 + (1-\omega)\,\mathrm{Var}_i\left[p_{i,t-1}\right],$$

where $\Delta_{e,t} \equiv \int_0^1 \left(\mathbb{E}_{i,t}\pi_t - \overline{\mathbb{E}}_t\pi_t\right)^2 di$ is cross-sectional forecast dispersion as in equation (17). Summing the geometric series in the discounted loss,

$$\sum_{t=0}^{\infty} \beta^t \mathrm{Var}_i\left[p_{i,t}\right] = \frac{1}{1-\beta(1-\omega)} \sum_{t=0}^{\infty} \beta^t \left[\frac{1-\omega}{\omega}\,\pi_t^2 + \omega\,\Delta_{e,t}\right] + \text{constant}.$$

Combining the output gap and price dispersion terms, and using $\mathrm{mc}_t = (\sigma+\psi)y_t + u_t$ (equation (19)), the discounted welfare function takes the form of (20) with

$$\lambda = \sigma + \psi, \qquad \gamma = \frac{\theta}{1-\beta(1-\omega)}, \qquad \vartheta = \omega\,(\theta-1)\,. \tag{22}$$

The structural primitives (σ, ψ, θ) therefore determine the reduced-form coefficients $(\lambda, \gamma, \vartheta)$. The welfare weight on output (λ) equals the sum of the inverse elasticities of intertemporal substitution and labor supply; the weight on price dispersion (γ) is increasing in the demand

elasticity and the discount factor; the weight on forecast accuracy (ϑ) is increasing in both the reset probability and the demand elasticity.

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